

- 6 (a) Define *magnetic flux*.

..... [1]

- (b) A rectangular coil has dimension 20 mm by 35 mm and 650 turns. The coil rotates about a horizontal axis perpendicular to a uniform magnetic field of flux density $2.5 \times 10^{-3} \text{ T}$, as shown in Fig. 6.1.

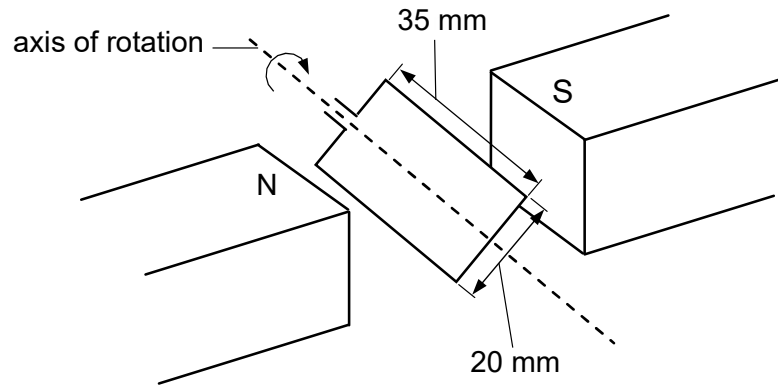


Fig. 6.1

The plane of the coil makes an angle θ with the horizontal, as shown in Fig. 6.1.

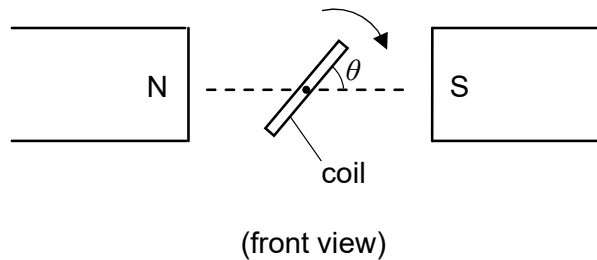


Fig. 6.1

The coil turns from $\theta = 30^\circ$ to $\theta = 0^\circ$ in 5.0 ms.

The coil has a resistance of 5.0Ω .

- (i) Calculate the magnetic flux linkage of the coil when $\theta = 30^\circ$. Give an appropriate unit.

magnetic flux linkage = [3]

- (ii) Calculate the average induced e.m.f. across the coil.

e.m.f. = V [2]

- (iii) Calculate the amount of charge flow through the coil in 5.0 ms.

charge = C [2]

[Total: 8]